7 a)

Kerckhoff's principle: the adversary knows every thing about the cryptosystem except the key(s).

b) An authentication system which requires two personal pieces of information is called "two factor".

c) in symmetric cryptography the deciphering key can be readily deduced from the enciphering key.

d) In Public key cryptography ...

... can not be .... .

e) In a block cipher the message is divided into blocks of equal length, and the enciphering function is applied to each block.

f) In a stream cipher the key is of infinite length and is just added to the plaintext to produce ciphertext.

g) The <u>key space</u> is the set of all possible keys.

h) A ciphertext only attack is one where no plaintext is available, i.e. frequency analysis, exhaustive search.

k) A computationally secure cipher is one where it is believed that it is safe with current computing power and methods.

i) $\mathbb{P}$ = plaintext space

$\mathbb{C}$ = ciphertext space

A cipher is perfectly secure

if

$$\text{Prob}(\text{Plain} = n \mid \text{Cipher} = m) = \text{Prob}(\text{Plain} = n)$$

for all $n \in \mathbb{P}$, $m \in \mathbb{C}$.

n) Public key signature with message recovery.

Signature = Alice's private key + message

Signature + Alice's public key = YES/NO + message

$f_E : (\mathbb{Z}_p)^d \to (\mathbb{Z}_p)^d,\ v \mapsto Av + B$

over an alphabet of p letters,

p a prime.

p^d choices for B.

Now A has to be invertible.

$$A = \begin{pmatrix} \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \end{pmatrix}$$

$\lambda R + \mu S$

$\lambda, \mu \in \mathbb{Z}_p$

$$(p^d - 1)(p^d - p)(p^d - p^2) \cdots (p^d - p^{d-1})$$

choices for first row

choices for 3rd row

Size of key space:

$$p^d \cdot (p^d - 1)(p^d - p)(p^d - p^2) \cdots (p^d - p^{d-1}).$$

$P = 29$

An exhaustive search can't be performed if the key space is of size $\geq 2^{80}$.

Q 10) Book work.